LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Dial indicators advantages, classifications, care and use of dial indicators
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Dial indicators advantages, classifications, care and use of dial indicators
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of dial indicator zeroing, spindle selection, and measurement on test block; Safety: Secure workpiece, avoid dropping indicator; Supervision: Instructor monitors zeroing technique and reading interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of Engineering Metrology?
(b) Name two common types of measuring instruments used in aviation maintenance.
(c) Why is precision measurement critical in aircraft engine assembly?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Advantages of dial indicators over vernier calipers • Classification by range (0-1mm, 0-10mm, 0-50mm) • Classification by mounting type (magnetic, anvil, stand) • Mechanism of gear train and magnification • Applications in runout and parallelism checks	L & Ds	PPT, IB	15 min
(b)	• Proper zeroing procedure and spindle selection • Reading dial faces and interpreting positive/negative values • Care and maintenance including cleaning and lubrication • Storage requirements to prevent damage • Common errors and how to avoid them	L & Ds	PPT, PDF	20 min
(c)	• Calibration verification procedures • Troubleshooting indicator drift or sticking • Recording measurement data in logbooks • Safety precautions during handling • Integration with other metrology tools	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a magnetic base and an anvil mount dial indicator.
(b) Describe the correct steps to zero a dial indicator before taking a measurement.
(c) List three essential care practices to maintain dial indicator accuracy.

12. Home Assignment and Follow Up	Prepare a brief report on the specific dial indicator range required for measuring the clearance in a typical aircraft piston engine cylinder, referencing the next lesson on comparators.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Comparators, introduction types and their application	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Lt. Cdr. Sharma, P No. 12345	Capt. Verma, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min